



SMEs prepare and revise interview question pool.



Conduct pilot testing with a small group.



Collect data using the developed personality structured interview.

Considering the transcript, please indicate the extent to which you agree or disagree on a scale from 1 to 5.



1. I am someone who is outgoing, sociable.
2. I am someone who is compassionate, has a soft heart.
3. I am someone who tends to be disorganized.
- ...

Interview-based **personality responses** simulation.

**Q:** To get us started, where are you from? where did you grow up and what was the place like?

**A:** Overland park, Kansas. It was a good neighborhood a safe place.

**Q:** Thinking back, what kind of student were you in school?



I will respond according to the transcripts.

PSID method (for more details and other methods, see Appendix E)

Interview-based **personality-related behaviors** simulation.

Based on the provided data, simulate the individual and answer the question. Recall that in the past year, how often have you done the following things?



1. Purposely wasted your employer's materials/supplies.
2. Complained about insignificant things at work.
3. Told people outside the job what a lousy place you work for.
- ...



Compare actual human response to simulated response.



Compare human sample with different methods LLM-simulated sample.



Explore whether LLMs exhibits personality-related behavior.

**BFI 2 Item 1:** I am someone who is outgoing, sociable.



4 = "agree a little".

3 = "neither agree nor disagree". While I engage in activities like working out and hanging out with friends, which are social in nature, my responses don't strongly emphasize a highly outgoing or sociable personality.



**Human BFI2 Scores #3**

**Human BFI2 Scores #2**

**Human BFI2 Scores #1**

|                    |      |
|--------------------|------|
| Extraversion:      | 2.75 |
| Agreeableness:     | 3.42 |
| Neuroticism:       | 3.00 |
| Conscientiousness: | 3.42 |
| Openness:          | 3.17 |
| ...                |      |

**Simulated BFI2 Scores #3**

**Simulated BFI2 Scores #2**

**Simulated BFI2 Scores #1**

|                    |      |
|--------------------|------|
| Extraversion:      | 3.22 |
| Agreeableness:     | 3.74 |
| Neuroticism:       | 3.07 |
| Conscientiousness: | 3.51 |
| Openness:          | 3.76 |
| ...                |      |



Can LLM simulate human personality distributions in terms of **descriptive statistics** and **factor analysis** results?

**CWB Item 1:** Purposely wasted your employer's materials/supplies.



1 = "never".

Question 1 Score: 1. I am unlikely to purposely waste materials or supplies, as I value responsibility and conscientiousness.

